M.Kumarasamy College of Engineering
NAAC Accredited Autonomous Institution
Approved by AICTE & Affiliated to Anna University
ISO 9001:2015 Certified Institution
Thalavapalayam, Karur - 639 113, TAMILNADU.

A Project Report

on

JOB PORTAL SYSTEM

Submitted in partial fulfillment of requirements for the award of the course

of

CGB1221-DATABASE MANAGEMENT SYSTEMS

Under the guidance of

Ms.A.Nithyasri

Assistant Professor / AI

Submitted By

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

M.KUMARASAMY COLLEGE OF ENGINEERING
(Autonomous)

KARUR – 639 113

MAY 2025

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M. KUMARASAMY COLLEGE OF ENGINEERING

(Autonomous Institution affiliated to Anna University, Chennai)

KARUR – 639 113

BONAFIDE CERTIFICATE

This is to certify that this project report on **"JOB PORTAL SYSTEM"** is the bonafide work of **ROOBA B (927623BAD092)** who carried out the project work during the academic year 2024 - 2025 under my supervision.

Signature

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

VISION OF THE INSTITUTION

To emerge as a leader among the top institutions in the field of technical education

MISSION OF THE INSTITUTION

- Produce smart technocrats with empirical knowledge who can surmount the global challenges
- Create a diverse, fully-engaged, learner-centric campus environment to provide quality education to the students
- Maintain mutually beneficial partnerships with our alumni, industry, and Professional associations

VISION OF THE DEPARTMENT

To excel in education, innovation, and research in Artificial Intelligence and Data Science to fulfil industrial demands and societal expectations.

MISSION OF THE DEPARTMENT

M1: To educate future engineers with solid fundamentals, continually improving teaching methods using modern tools.

M2: To collaborate with industry and offer top-notch facilities in a conducive learning environment.

M3: To foster skilled engineers and ethical innovation in AI and Data Science for global recognition and impactful research.

M4: To tackle the societal challenge of producing capable professionals by instilling employability skills and human values.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

Graduates will be able to:

- **PEO 1:** Compete on a global scale for a professional career in Artificial Intelligence and Data Science.
- **PEO 2:** Provide industry-specific solutions for the society with effective communication and ethics.
- **PEO 3:** Hone their professional skills through research and lifelong learning initiatives.

PROGRAM OUTCOMES (POs)

Engineering students will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, give and receive clear instructions.

11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

- **PSO1:** Capable of finding the important factors in large datasets, simplify the data, and improve predictive model accuracy.
- **PSO2:** Capable of analyzing and providing a solution to a given real-world problem by designing an effective program.

ABSTRACT

A Job Portal System is a web-based application designed to streamline the job search and recruitment process using a Database Management System (DBMS). The system facilitates job seekers in searching and applying for relevant job opportunities while enabling employers to post vacancies and manage applications efficiently. The DBMS stores and organizes job listings, candidate profiles, resumes, and application statuses, ensuring easy retrieval and management of data. For job seekers, the platform provides a searchable job database, profile creation, and application tracking, enhancing their job-hunting experience. Employers can post job openings, review applications, and communicate with candidates through an intuitive interface. The system ensures data integrity, security, and efficient query processing, offering a seamless experience for both candidates and recruiters. With advanced search and filtering mechanisms, the platform enhances the recruitment process by matching the right candidates with suitable job opportunities, optimizing hiring workflows.

KEYWORDS: *Job Portal System, Database Management System (DBMS), recruitment, job seekers, employers, job listings, candidate resumes, application statuses, profile creation, application tracking, job openings, search and filtering,*

ABSTRACT WITH POs AND PSOs MAPPING

ABSTRACT	COs MAPPED	POs MAPPED	PSOs MAPPED
A Job Portal System is a web-based application designed to streamline the job search and recruitment process using a Database Management System (DBMS). The system facilitates job seekers in searching and applying for relevant job opportunities while enabling employers to post vacancies and manage applications efficiently. The DBMS stores and organizes job listings, candidate profiles, resumes, and application statuses, ensuring easy retrieval and management of data. For job seekers, the platform provides a searchable job database, profile creation, and application tracking, enhancing their job-hunting experience. Employers can post job openings, review applications, and communicate with candidates through an intuitive interface. The system ensures data integrity, security, and efficient query processing, offering a seamless experience for both candidates and recruiters. With advanced search and filtering mechanisms, the platform enhances the recruitment process by matching the right candidates with suitable job opportunities, optimizing hiring workflows.	CO1 CO2 CO3	PO 1 PO 2 PO 3 PO 5 PO 6 PO 7 PO 8 PO 9 PO 10 PO 11 PO 12	PSO1 PSO2

Note: 1- Low, 2-Medium, 3- High

SUPERVISOR **HEAD OF THE DEPARTMENT**

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INTRODUCTION

1.1 PROBLEM STATEMENT

Design and develop a Job Portal System using a Database Management System (DBMS) to store job listings, candidate profiles, and application statuses. The system should allow job seekers to search and apply for jobs, while employers can post vacancies and manage applications. The DBMS should efficiently manage job data, application processes, and candidate resumes, providing both parties with a seamless experience.

1.2 INTRODUCTION

In today's digital era, job portals have become an essential bridge between job seekers and employers , revolutionizing the recruitment process. A Job Portal System is a web-based application designed to simplify and optimize job searching and hiring through the efficient use of a Database Management System (DBMS). By leveraging a well-structured database, the system enables job seekers to explore relevant opportunities, create profiles, and track applications while allowing employers to post vacancies, filter candidates, and manage recruitment seamlessly. With an intuitive interface and robust data management capabilities, the Job Portal System enhances user experience by ensuring efficient query processing, data security, and integrity. The platform provides a centralized database for storing job listings, candidate details, resumes, and application statuses, making it easy to access and retrieve essential information.

CHAPTER 1

1.3 OBJECTIVE

The primary objective of this project is to design and develop a comprehensive Job Portal System utilizing a Database Management System (DBMS) to effectively store and manage job listings, candidate profiles, and application statuses. The system aims to streamline the job search and recruitment process by providing job seekers with a platform to explore opportunities, submit resumes, and apply for suitable positions. Simultaneously, it enables employers to post job vacancies, review candidate applications, and manage the recruitment workflow efficiently. The DBMS will play a critical role in ensuring data integrity, secure storage, and efficient retrieval of information, thereby offering a seamless and user-friendly experience for both job seekers and employers.

1.4 DBMS

In this Job Portal System, the DBMS is used to efficiently store and manage data related to job listings, users, and applications. It organizes information in structured tables with relationships among candidates, employers, and job posts. Job seekers can retrieve job data using search filters, and employers can track applications in real time. The DBMS ensures data integrity and security throughout the system. It supports efficient queries for matching candidates with job requirements. Overall, it enables smooth, reliable, and scalable operations for both job seekers and employers.

1.5 PHP

PHP is effectively used in this Job Portal System to handle server-side operations and dynamic content. It manages user registration, login, and logout processes through files like register.php and login.php. PHP handles job postings and applications using scripts such as post_job.php, apply.php and application_status.php. It connects to the database using db.php and config.php to store and retrieve data securely. Dashboards for job seekers and employers are powered by PHP to show personalized content. Passwords are secured using hashing methods implemented in hash_generator.php. Overall, PHP ensures smooth interaction between the user interface and backend database operations.

PROJECT METHODOLOGY

2.1 PROPOSED WORK

Jobseeker Workflow:

1. **Register/Login:** Jobseeker can register or log in using their credentials to access the system.
2. **Filter & search the job:** Using filter option jobseeker can search the jobs based on category, location, salary, company.
3. **Apply for the job:** After filtering the jobs they can apply for the job by filling the resume and skills.
4. **Profile creation:** Jobseeker can apply for the job by filling their resume, skills and experience.
5. **Application status:** Jobseeker can view their application status whether it is accepted, pending or confirmed with applied date.

Employer Workflow:

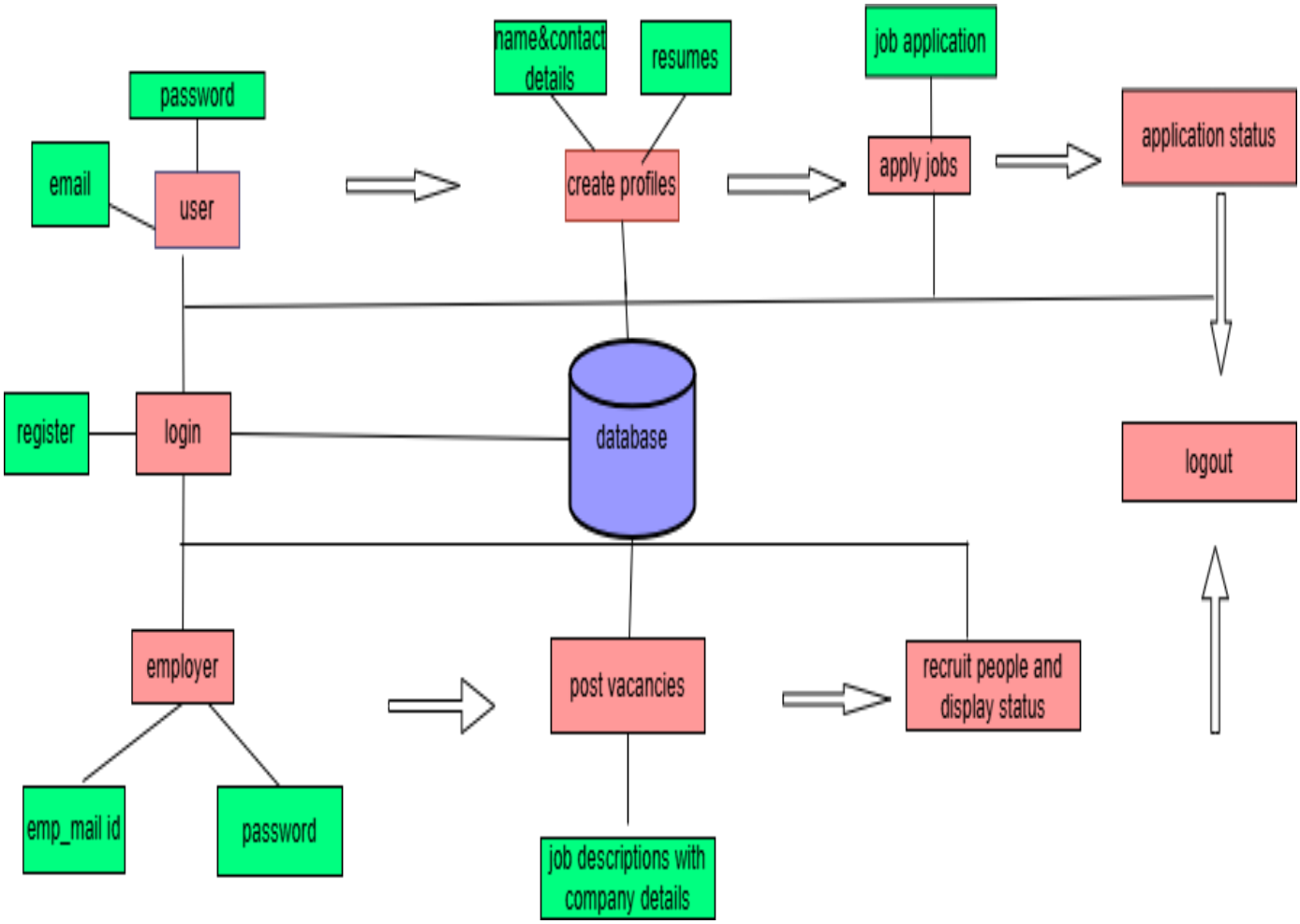
1. **Register/Login:** Employer log in or register with employer credentials with company details to access the system.
2. **Post the job:** Employers can post the job which is needed for their company with credentials like job category, location, salary and its description.
3. **View application profiles:** Employer can view the application profiles for a posted job with candidate name, resume, skills and experience for the posted

CHAPTER 2

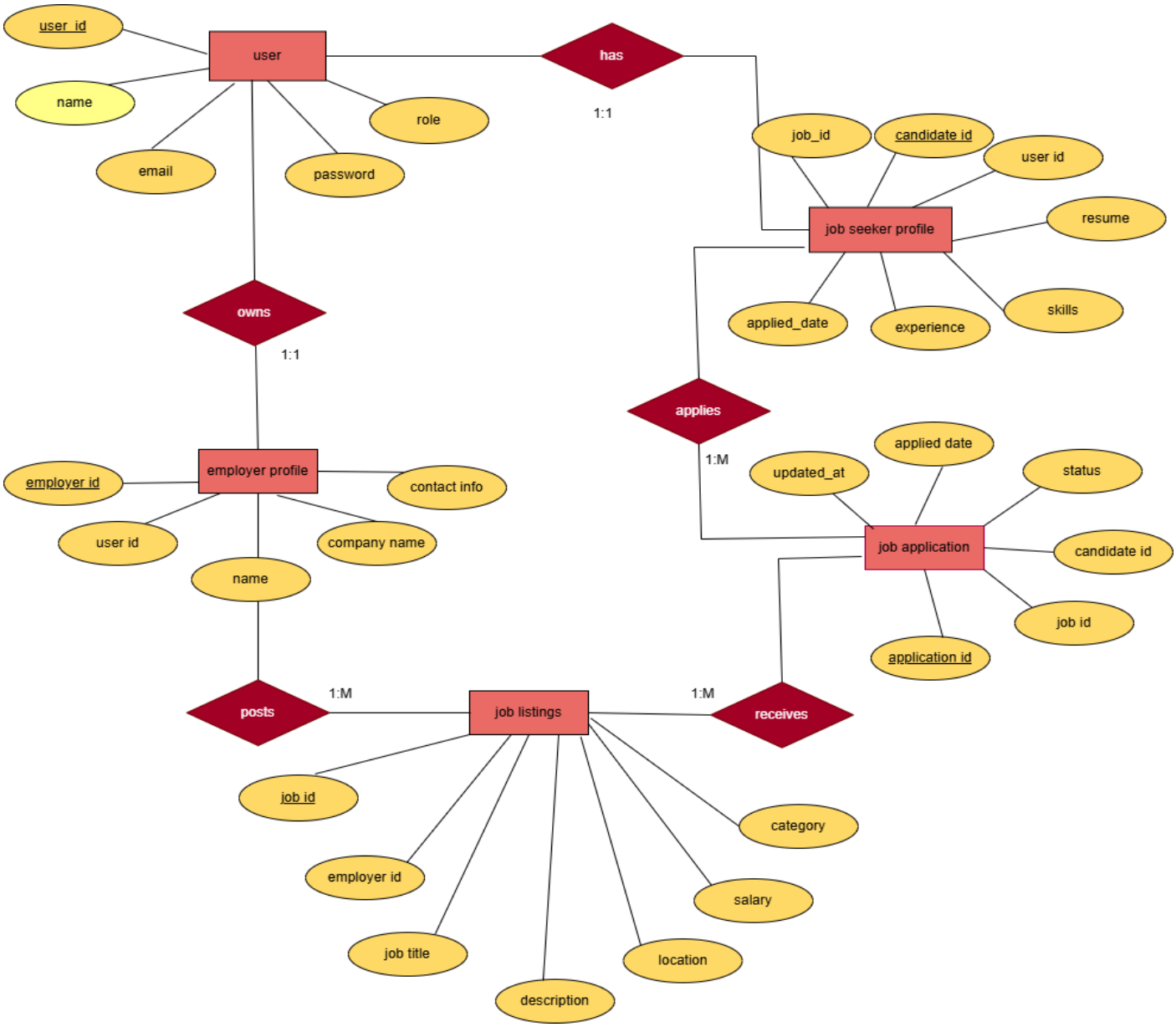
specific job.

4. **Profile selection:** After verifying their profiles employers can accept or reject the profiles for the job.
5. **View the status of profiles:** Employers can also view the status like accept, reject or pending profiles of jobseeker. It confirms the status of applicants for a job.

2.2 ARCHITECTURE



2.3 E-R DIAGRAM



2.4 SCHEMA DIAGRAM

Users Table

User_id	name	Email_id	Password	Role
1	Baskaran M	Baskaran@gmail.com	$2y$10$5v5....	Employer

Employer profile Table

Employer _id	name	User_id	Company	Contact_info
91	Baskaran M	1	Tata Consultancy Services (TCS)	Chennai, India, contact no: 3547576354

Jobseeker_profile Table

User_id	Candidate_id	Job_id	Resume	Skills	Experience	Applied_date
5	19	2	i completed my schooling...	have experience in powerbi ai....	1 year	2025-04-13

Job_listings Table

Employer_id	Job_id	Job_title	Description	Location	Salary	Category
91	2	graphic designer	knowledge in photoshop....	Chennai	30000	designer

Job application Table

Application _id	Job _id	Candidate_id	Applied_date	Updated_at	Status
5	2	19	2025-04-13	2025-05-10 12:00:12	Rejected

CHAPTER 3
SOFTWARE REQUIREMENTS

3.1 Front End

In the job portal system, HTML and CSS are used exclusively to build and style the front end, which is the part of the website users interact with directly. HTML (Hyper Text Markup Language) is used to create the structure and layout of the web pages—defining elements like forms, input fields, buttons, tables, and navigation menus. For example, in files like register_form(html).php, login(html).php, and post_job(html).php, HTML is responsible for creating the registration form, login interface, and job posting form respectively. On the other hand, CSS (Cascading Style Sheets) is used to control the presentation and visual design of these HTML elements. It defines styles such as colors, fonts, spacing, alignment, borders, button styles, and responsiveness, ensuring the interface looks clean, professional, and consistent across all pages. Together, HTML and CSS provide a smooth and user-friendly experience, making it easy for job seekers and employers to navigate, view content, and interact with the system.

3.1 Back End

In the job portal system, PHP and SQL (MySQL) are used as the backend technologies to handle the logic, data processing, and database operations. Running through XAMPP, PHP acts as the server-side scripting language that manages tasks such as user registration, login authentication, job posting, application submissions, and dashboard updates. It connects with a MySQL database, where all the essential data is stored and managed. Using phpMyAdmin, i have created and managed several database tables such as `users`, `employer profile`, `job_listings`, `jobseeker_profile`, and `job application`. These tables store user details, employer information, job posts, job seeker resumes, and application statuses, respectively. PHP scripts interact with these tables using SQL queries to insert, retrieve, update, or delete data based on user actions, ensuring dynamic and data-driven functionality throughout the platform. This backend setup enables seamless communication between the user interface and the database, making the job portal fully functional and interactive.

CHAPTER 4
MODULE DESCRIPTION

4.1 Job seeker Module :

The **Job Seeker** module enables users to register and log in to the job portal system to actively participate in the job search process. Once logged in, job seekers can browse available job listings and apply for jobs that match their interests and qualifications. The job application includes essential information such as the candidate's resume, key skills, and work experience. After applying, job seekers can track the status of their applications in real time, with updates indicating whether the application is **Accepted**, **Rejected**, or still **Pending**. This module ensures a streamlined and user-friendly experience for candidates navigating their career opportunities through the portal.

4.2 Employer Module

The **Employer** module allows company representatives to register and log in to the job portal system to manage recruitment activities for their respective organizations. Once authenticated, employers can post job openings with relevant details under their company's name. They can view applications submitted by job seekers for each job listing, including candidate resumes, skills, and experience. Employers have the ability to review and update the application status by accepting, rejecting, or marking applications as pending. This module provides employers with an efficient platform to manage job postings and streamline their hiring process.

4.3 Application Status Module

The **Application Status** module serves as the central interface for managing

and tracking job applications submitted through the portal. It bridges the interaction between job seekers and employers by displaying the real-time status of each job application. Job seekers can view the current status of their applications—whether **Accepted**, **Rejected**, or **Pending**—for every job they have applied to. Simultaneously, employers can access and manage the list of applicants for each job posting, review their profiles, and update the status of each application accordingly. This module ensures transparency and seamless communication throughout the recruitment process.

CHAPTER 5
IMPLEMENTATION

5.1 SOURCE CODE

LOGIN.HTML

```
prepare("SELECT user_id,password, role FROM users WHERE email_id = ?");
if (!$stmt) { $error = "Prepare failed: " . $conn->error; }
 else {
 $stmt->bind_param("s", $email_id); $stmt->execute();
 $result = $stmt->get_result();
if ($result->num_rows === 1) {
 $row = $result->fetch_assoc();
 $role = strtolower(trim($row['role']));
if (password_verify($password, $row['password']))
 {
 $_SESSION['user_id'] = $row['user_id'];
$_SESSION['role'] = ucfirst($role);
if ($role === "employer")
 {header("Location: employer_dashboard.php"); exit; }
elseif ($role === "job seeker")
{ header("Location: jobseeker_dashboard.php"); exit; }
 else { $error = "Unknown role: " . htmlspecialchars($role); } }
else { $error = "Invalid password."; } }
else { $error = "No account found."; }
$stmt->close(); } }
 $conn->close();
}
?>
```

```
<!DOCTYPE html>
<html>
<head>
   <title>Login</title>
   <style>
      body {
         margin: 0;
         padding: 0;
         font-family:
'SegoeUI', Tahoma, Geneva, Verdana, sans-serif;
         display: flex;
         justify-content: center;
         align-items: center;
         height: 100vh;
         position: relative;
         overflow: hidden; }
      body::before {
         content: "";
         position: absolute;
         top: 0;
         left: 0;
         height: 100%;
         width: 100%;
         background-image: url('https://img.freepik.com/premium-vector/log-people-isometric-
vector-illustration-illustration-with-sign-people-mobile-app-design_123447-4034.jpg?w=996');
         background-size:
cover;
         background-position: center;
```

CHAPTER 5

```
    filter: blur(3px);
    opacity: 1.5;
    z-index: -1;}
 .login-container {
    background-color: rgba(255, 255, 255, 0.95);
    padding: 40px;
    border-radius:12px;
    box-shadow: 0px 4px 15px rgba(0, 0, 0, 0.2);
    width: 350px;
    z-index: 1;}
 h2 {
    text-align: center;
    color: #ff5e62;
    margin-bottom:25px:}
 label {
    font-weight: bold;
    display: block;
    margin-top: 15px;
    color: #333;}
 input[type="email"],
input[type="password{
    width: 100%;
    padding: 10px;
    margin-top: 5px;
    border: 1px solid #ccc;
    border-radius: 6px;
    font-size: 16px; }
 button {
```

```
            width: 100%;
            margin-top: 25px;
            padding: 12px;
            background-color: #ff5e62;
            color: white;
            border: none;
            font-size: 18px;
            border-radius: 8px;
            cursor: pointer;
            transition: 0.3s
ease:}
        button:hover {
            background-color: #e94e50; }
        .error {
            color: red;
            text-align: center;
            margin-top: 10px: }
    </style>
</head>
<body>
<div class="login-container">
    <h2>Login</h2>
    <?php if
(!empty($error)): ?>
        <div
class="error"><?php echo
$error; ?></div>
    <?php endif; ?>
    <form
method="POST">
```

CHAPTER 5

```
        <label>Email:</label>

        <input type="email"
name="email_id"
required>


<label>Password:</label>

        <input
type="password"
name="password"
required>

        <button
type="submit">Login</but
ton>

    </form>

</div>

</body>

</html>
```

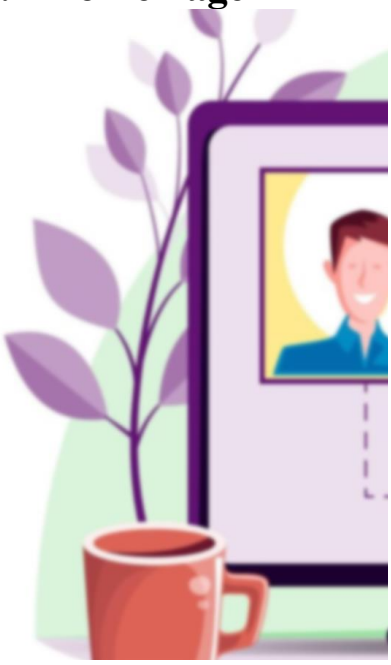
LOGIN.PHP

```
prepare("SELECT user_id, password, role FROM users WHERE email_id = ?");
 if (!$stmt) { $error = "Prepare failed: " . $conn->error; }
 else {
$stmt->bind_param("s", $email_id);
$stmt->execute(); $result = $stmt->get_result();
 if ($result->num_rows === 1) {
 $row = $result->fetch_assoc();
$role = strtolower(trim($row['role']));
 // Store role as lowercase
 if (password_verify($password, $row['password'])) {
 $_SESSION['user_id'] = $row['user_id'];
 $_SESSION['role'] = $role;
// Store role as lowercase
if ($role === "employer")
{ header("Location: employer_dashboard.php"); exit; }
 elseif ($role === "job seeker")
 { header("Location: jobseeker_dashboard.php"); exit; }
 else { $error = "Unknown role: " . htmlspecialchars($role); } }
 else { $error = "Invalid password."; } }
 else { $error = "No account found."; }
 $stmt->close(); }
} $conn->close();
 }
 ?>
```

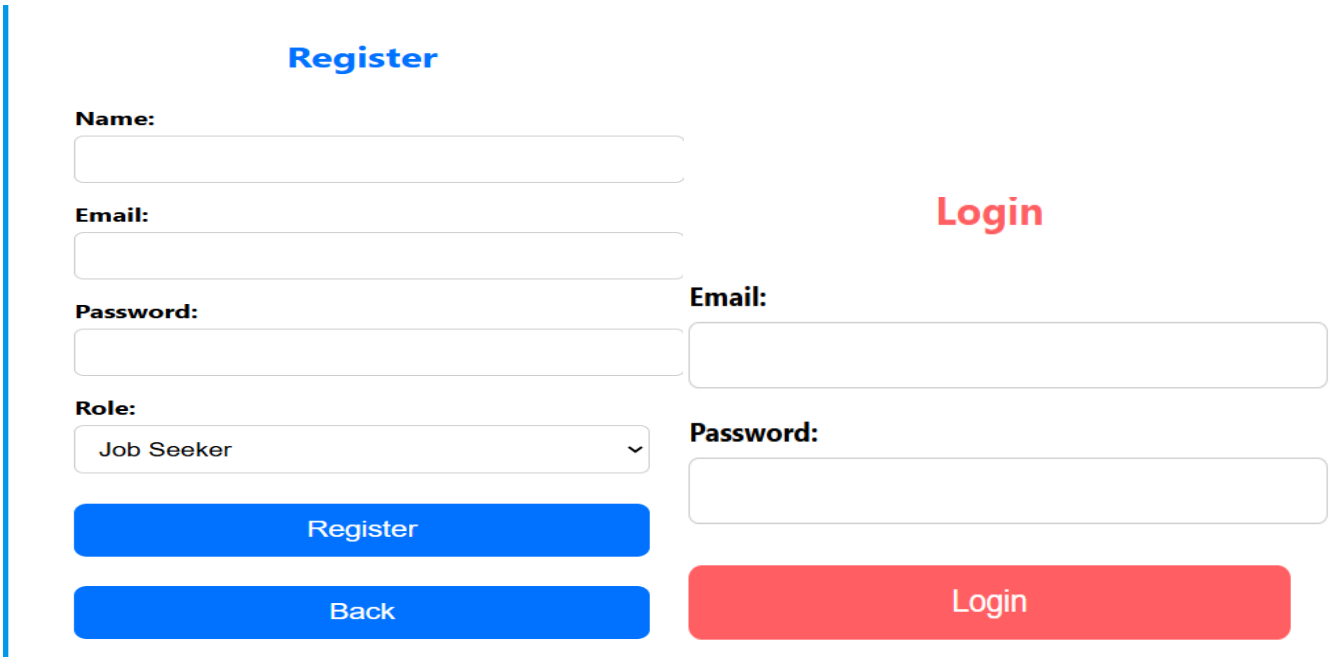
CHAPTER 6

SNAPSHOTS

6.1 Home Page



6.2 Login/Register Page



6.3 Jobseeker Dashboard Page

Welcome to Job Seeker Dashboard

Application Forms | Logout

Company: Tata Consultancy Service(TCS)
Job Title: graphic designer
Description: knowledge in adobe photoshop, adobe illustrator, figma, adobe indesign and with 1 year of experience
Location: chennai
Salary: $30000
Category: designer
Apply Job

Company: Tata Consultancy Service(TCS)
Job Title: web development
Description: Frontend: A web developer should know HTML to structure web pages properly. Backend: Understanding PHP helps in building dynamic and interactive websites. Database: Knowledge of MySQL is essential for storing and retrieving data.
Location: bangalore
Salary: $400000
Category: developer
Apply Job

6.4 Applying job Page

Apply for Job

Resume:

Skills:

Experience:

Submit Application

Back to Dashboard

6.5 Employer Dashboard page

loyer

Tata Consultancy Service(TCS)

Welcome, Baskaran!

Post New Vacancies | View Application Profiles | Manage Application Status

6.5 Post the Job Page

Job Title:

Description:

Location:

Salary:

Category:

Post Job

Job Listings

6.6 Application Profiles Dashboard Page

Current role: Employer

Application Profiles

Logout

Back to Dashboard

Candidate Name	Job Title	Resume	Skills	Experience	Applied Date	Status
sadhana	graphic designer	Completed high school with a focus on science, achieving strong academic performance. Graduated with a Bachelor's degree in Computer Science from Anna University , specializing in software development.	Proficient in Adobe Creative Suite (Photoshop, Illustrator, InDesign) and UI/UX design principles. Skilled in creating visually engaging graphics, branding, and digital media for diverse projects.	3 years	2025-05-03	Rejected Update
ROOBA B	web development	Completed schooling from Sri sankara vidhyalaiya, achieving strong academic results. Graduated from Annai magalir college with a degree in information technology(IT)/B.Tech Skilled in HTML, CSS, JavaScript, PHP, and MySQL with hands-on web development ex	Frontend: HTML, CSS, JavaScript, React, Vue.js, Backend: PHP, Node.js, Python, Ruby, Databases: MySQL, MongoDB, PostgreSQL Version Control: Git, GitHub, API: REST, GraphQL	2 years	2025-04-18	Rejected Update
ROOBA B	graphic designer	i completed my schooling in meenakshi matric school and college in kumarasamy in karur and i am working as a data scientist in tcs	i have experience dealing with powerbi,ai tools and scientific analysis tools and ai basic toools	1 year	2025-04-13	Pending Update

6.7 Application Status page

Current role: Job seeker

Job Portal System Logout

Your Job Applications

Candidate Name	Company	Job Title	Applied Date	Status
ROOBA B	Tata Consultancy Service(TCS)	web development	2025-04-18	Rejected
ROOBA B	Tata Consultancy Service(TCS)	graphic designer	2025-04-13	Pending

CONCLUSION

The Job Portal System developed in this project is a dynamic web-based application designed to streamline the interaction between job seekers and employers. The **frontend** of the system is created using **HTML** and **CSS**, offering user-friendly interfaces for logging in, applying to jobs, and managing employer/job seeker dashboards. The **backend** is built with **PHP**, which handles the core logic such as session control, form handling, and server-side validations. All data storage and retrieval operations are performed using **MySQL**, a relational database system, with SQL queries managing tables like users, job_listings, employer profile, jobseeker_profile, and job application. The system supports role-based access, allowing employers to post jobs and view applications, while job seekers can apply for jobs and track their application status. The use of secure login with hashed passwords enhances data integrity and user privacy. Overall, this project demonstrates an effective integration of frontend and backend technologies to deliver a functional, secure, and responsive job portal system for real world application.

REFERENCES

1.Silberschatz, A., Korth, H.F., & Sudarshan, S. (2019). *Database System Concepts* (7th ed.). McGraw-Hill Education – a foundational book covering all DBMS principles used in building job portals.

2.NevonProjects. *Job Portal Project Report*. Available at: https://nevonprojects.com/job-portal-project – provides a practical job portal.